Microsoft Fabric

Enterprise analytics with Copilot, OneLake and Fabric IQ

OneLake foundations | Copilot in Fabric and Power BI | Fabric IQ ontology | Fabric data agents

One lake, one copy

Questions in plain language

Governed from end to end

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Five Days from Cloud Basics to Production-Ready AI

Each day adds one layer. Every day mixes short concepts, a live demo and a hands-on lab.

DAY 1

Azure & Agent Foundations

Agentic workflows with Microsoft 365 Copilot

- Azure cloud foundations
- How AI agents work
- Agent Builder, Notebooks, Cowork, Skills
- Chaining agents with a human in the loop

DAY 2

Copilot Studio

Low-code and pro-code conversational AI

- Agents, workflows and apps
- Knowledge, MCP tools, skills
- Topics and the standard harness
- Evaluate, publish, govern

DAY 3

Microsoft Foundry

Azure AI foundation and core concepts

- Projects, resources, endpoints
- Model catalog and comparison
- Prompting and evaluation
- Secure, private, observable agents

DAY 4

Data, RAG & Power BI

Grounding AI in trusted data

- Data quality for AI
- Retrieval-augmented generation
- Azure AI Search patterns
- Power BI basics and dashboards

DAY 5 · TODAY

Microsoft Fabric

Enterprise analytics with Copilot

- OneLake, lakehouse, shortcuts
- Copilot in Fabric and Power BI
- Fabric IQ ontology and data agents
- Governance and the road ahead

Foundations → Microsoft 365 Copilot → Copilot Studio → Foundry → RAG & Power BI → Fabric

Today: From CSV Files to Questions in Plain English

Three labs built around one refinery reliability scenario.

01 OneLake foundations → 02 Copilot in Fabric → 03 Fabric IQ → 04 Bring it together

Lab 1 · ~45 min

- Two lakehouses
- Load refinery data into tables
- Shortcut: use data without copying
- SQL and the OneLake catalog

Lab 2 · ~50 min

- Copilot in a notebook
- Build a semantic model
- Copilot builds a Power BI page

Lab 3 · ~50 min

- Create a refinery ontology
- Bind entities to data
- Ask a Fabric data agent

Wrap-up

- Governance and capacity
- SATORP use-case roadmap
- Quality gate for go-live

You need

- Workspace on paid capacity F2+ (or P1+)
- Contributor role or higher
- Copilot and data agent tenant settings on
- Ontology (preview) setting for Lab 3
- The five CSV files

Trial capacities don't support Copilot. Check your capacity before Lab 2.

01

Microsoft Fabric and OneLake

One platform, one lake, one copy of the data

What Fabric is | Workloads | OneLake | Lakehouse & shortcuts | Medallion

What Is Microsoft Fabric?

An end-to-end analytics platform delivered as software as a service.

Data integration, engineering, warehousing, real-time analytics, data science and Power BI, in one place, on one copy of the data.

- **SaaS:** no servers or clusters to manage; you buy capacity.
- **One lake:** every item stores its data in OneLake.
- **Open format:** tables are Delta Parquet, readable by every engine.
- **Copilot built in:** in notebooks, Data Factory and Power BI.
- **Governed:** Entra ID, workspace roles, Microsoft Purview.

WHY IT MATTERS FOR AI

- Agents need clean, governed, current data
- One copy avoids conflicting numbers
- Data agents answer from curated tables

Fabric is where structured data becomes AI-ready.

The Fabric Workloads

Different experiences for different jobs, all sharing OneLake.

01 Data Factory

Pipelines and Dataflow Gen2 to ingest and transform data from hundreds of sources.

02 Data Engineering

Lakehouses and Spark notebooks for large-scale preparation.

03 Data Warehouse

Full SQL warehouse for structured, relational analytics.

04 Real-Time Intelligence

Eventstreams, Eventhouse and KQL for streaming sensor and event data.

05 Data Science

Notebooks, ML models and Fabric data agents.

06 Power BI

Semantic models, reports and dashboards, with Copilot.

Plus Fabric IQ: the business-meaning layer of ontologies and agents on top of OneLake.

OneLake: OneDrive for Data

One organisation-wide data lake that comes with every Fabric tenant.

One lake	One copy	Open format
One governed place	**Shortcuts instead of copies**	**Delta Parquet tables**
• All Fabric data in one lake • Not dozens of copies in different tools • Organised by workspaces and items	• Reuse data where it already lives • OneLake, ADLS Gen2, Amazon S3, Google Cloud Storage, Dataverse and more • No duplication, no drift	• Spark, SQL, Power BI and KQL engines read the same table • No exporting between tools • Open, not locked in

Three ideas to remember from Lab 1: one lake, one copy, open format.

Lakehouse, Warehouse or Eventhouse?

All three store data in OneLake. Choose by the shape of the data and the skills of the team.

Item	Best for	Main language	Refinery example
Lakehouse	Files and tables together; flexible preparation	Spark (Python), SQL to read	Production and downtime CSVs (Lab 1)
Warehouse	Structured, relational analytics with full SQL	T-SQL	Finance and planning models
Eventhouse	High-volume streaming and time-series data	KQL	Sensor and historian readings
Mirrored database	A near real-time replica of an operational database	SQL	Copies of operational systems for analytics

Every lakehouse also gets a SQL analytics endpoint, so SQL users can query it without Spark.

The Medallion Pattern: Bronze, Silver, Gold

A common way to organise lakehouse data from raw to ready.



Day 4 connection

The cleaning you did yesterday is the bronze-to-silver step.

For AI

Point data agents at gold tables with clear names and descriptions, never at raw data.

Anti-pattern

“A Fabric data agent over raw, unnamed tables.” Curate first.

Lab 1: Two Lakehouses and a Shortcut

Bring refinery data into OneLake, then let the reliability team use it without copying it.

Workspace	Your workspace on F2+ capacity (5 min)
Refinery_LH	Upload the five CSVs and load them into tables (15 min)
Reliability_LH	A second lakehouse for the reliability team (5 min)
Shortcut	Point Reliability_LH at Refinery_LH tables: one copy, no duplication (10 min)
Query & find	SQL analytics endpoint queries, then the OneLake catalog (15 min)

Lab data (fictional)	
units.csv	6 process units with design capacity
equipment.csv	30 items: type, criticality A/B/C
production_daily	180 rows, actual vs planned, Sept 2026
downtime_events	37 events, July–Sept 2026
work_orders.csv	60 orders: priority, status, cost (SAR)

02

Copilot in Fabric

From data to insight in plain English

- Copilot in notebooks
- Semantic model
- Copilot in Power BI
- Prompts that work

Two Places You Use Copilot in Lab 2

Copilot helps builders write code and reports faster. You stay responsible for checking it.

Copilot in notebooks	Semantic model	Copilot in Power BI
Data engineering	**The business layer in between**	**Reporting**
• Writes PySpark and SQL from your prompts • Explains code and fixes errors • Plots charts and writes results to tables	• Measures and definitions • What Power BI, Copilot and Fabric IQ build on • Clear names = better Copilot answers	• Builds report pages from a description • Summarises insights from a semantic model • Answers questions about the report

Copilot output varies from run to run. Always read the code or visual before trusting it.

Prompts That Turn Data into Insight

Specific prompts name the table, the calculation, the filter and the output.

What makes these prompts work

- **Name the table** and the columns.
- **Define the calculation:** actual ÷ planned.
- **Filter:** unit, month, planned = 'No'.
- **Say what to do** with the result: sort, plot, save.

Notebook and Power BI prompts from Lab 2

```
1. Load the production_daily table and show the
   first 10 rows and the column data types.

2. Calculate throughput attainment percent
   (actual divided by planned) per UnitId for
   September 2026, join with units to show
   UnitName, and sort from lowest to highest.

3. From downtime_events, show unplanned events
   (Planned = 'No') for UnitId U400 in Sept 2026.

4. Write the attainment result to a new Delta
   table called unit_attainment.

Power BI: Create a report page for refinery
reliability: Attainment % by unit name, Unplanned
Downtime (h) by cause, Open Work Orders by priority.
```

Before Copilot Works: Capacity and Tenant Settings

Ask your Fabric admin to confirm these. Most Lab 2 problems start here.

01 **Paid capacity**

F2 or higher, or Power BI Premium P1+. Trial capacities don't support Copilot.

02 **Copilot tenant setting**

Users can use Copilot and other features powered by Azure OpenAI.

03 **Cross-region processing**

Only if your capacity region requires it. Decide with your compliance team.

04 **Workspace role**

Contributor or higher on the lab workspace.

DATA RESIDENCY CHECK

Day 1 lesson, applied:

- Where is the capacity?
- Where would prompts be processed?
- Is that approved for this data?

Compliance decides the cross-region setting, not the lab.

03

Fabric IQ: Ontology and Data Agents

Teach Fabric the language of the refinery

- Why ontology
- Entities & relationships
- Binding to data
- Ontology agent
- Data agent

Tables Speak Data. Engineers Speak Refinery.

Fabric IQ adds business meaning on top of OneLake so AI can answer in business language.

“This FCC compressor has two open P1 work orders and caused 40 hours of downtime.”

- **A data engineer sees:** equipment, work_orders, downtime_events tables.
- **A reliability engineer thinks:** units, equipment, work orders, downtime, and how they connect.
- **An ontology bridges the two:** entity types, properties and relationships, bound to real data.
- **Defined once, centrally:** every agent uses the same meaning of “unit” or “open”.

STATUS, OCTOBER 2026

- **OneLake, lakehouse, shortcuts:** GA
- **Fabric data agent:** GA
- **Copilot in Fabric:** F2+; some parts preview
- **Fabric IQ ontology:** preview

Preview labels and steps may change.

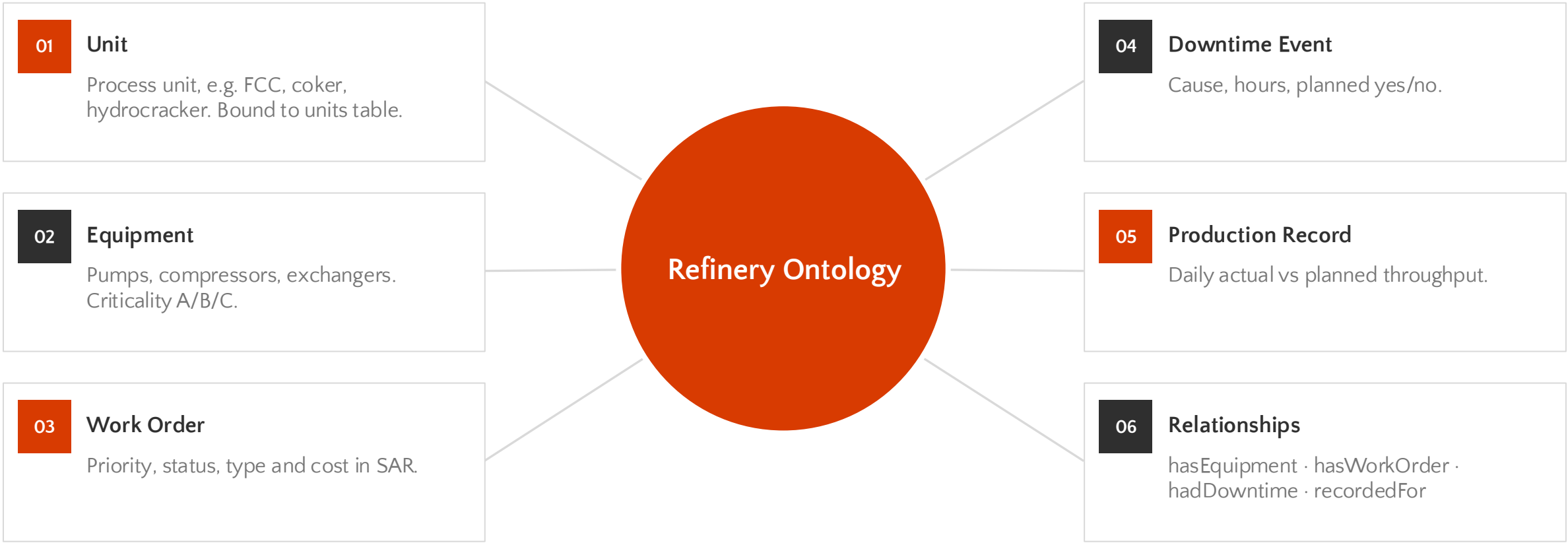
Fabric IQ in One Table

How the pieces you built all week fit together.

Fabric IQ part	What it is	In today's labs
OneLake	The unified data foundation	Lab 1 lakehouses
Semantic model	Measures and business definitions for BI	Lab 2 model
Ontology (preview)	Business entities, relationships, rules and metrics bound to data	Built in Lab 3
Graph	Optional execution layer to explore connected entities	Optional look
Data agent	Natural-language Q&A over your sources	Built in Lab 3
Operations agent	Monitors data and acts on conditions (Real-Time Intelligence)	Discussed only

The Refinery Ontology

Entity types bound to lakehouse tables, connected by named relationships.



The Refinery Reliability Data Agent

Add the ontology as a source, give clear instructions, then check the query it generates.

Building it

- **New item → Data agent:** Refinery Reliability Agent.
- **Add data:** RefineryOntology, or the five lakehouse tables.
- **Agent instructions:** define the business terms.
- **Check:** open the answer details to see the generated query.
- **Publish** so people and other agents can use it.

Agent instructions from Lab 3

```
You help SATORP reliability and operations teams.
- A "unit" is a process unit (for example the
  FCC, coker or hydrocrackers). Always show the
  unit name, not only the ID.
- "Unplanned downtime" means downtime events
  where Planned = 'No'.
- "Open" work orders are any work orders whose
  Status is not 'Completed'.
- Attainment % = actual throughput divided by
  planned throughput.
- Answer with a short table, then one sentence
  of explanation. Say clearly when the data does
  not contain the answer.
```

Test Questions with an Answer Key

Compare the agent's answers with known results from the fictional lab data.

Question	Expected answer	What it tests
Which unit had the most unplanned downtime, July–Sept 2026?	Fluid Catalytic Cracker, ~192.5 h	The meaning of "unplanned"
What was the top cause of unplanned downtime?	Fouling, ~184.5 h	Grouping and ranking
How many open or in-progress P1 work orders are on criticality A equipment?	3	Following relationships
What is the refinery's profit margin?	Says the data doesn't contain it	Refusing to guess

The last question matters most: a good data agent says "I don't have that" instead of inventing a number.

Fabric Data Agents: Strengths and Limits

From the SATORP decision guide. Plan for the limits before users meet them.

Use it when

"How much, how many, which, trend?"

- Structured data in OneLake
- Governed, read-only Q&A on curated data
- Up to five sources per agent: lakehouse, warehouse, semantic model, KQL, mirrored DB, ontology

Limits to plan for

Not a document or action agent

- Read-only: no writing data or actions
- No unstructured files
- English only
- Results capped at 25 rows × 25 columns
- Capacity F2+ (or P1+)

Make it good

Curate, instruct, test

- Clear table and column names and descriptions
- Agent and data-source instructions
- Example queries for tricky metrics
- A ground-truth question set re-run after changes

How the Platforms Work Together at SATORP

Layers that combine, not competitors. Build once, govern once, reuse everywhere.

Users	Employees in Teams and Microsoft 365 Copilot; website visitors
Copilot Studio	The front door: Employee Assistant, Public Q&A agent, workflows and approvals
Fabric data agent	KPIs, maintenance and production numbers from OneLake
Foundry agent	Engineering agent with private access to SAP and OpenText
SATORP systems	SAP S/4HANA (mirrored to OneLake), OpenText, SharePoint SOPs

Integration patterns	
Front door	One agent in Teams delegates to others
Fabric everywhere	Same data agent in Studio, Foundry, M365
Foundry in Teams	Connected agent or Agents SDK
MCP standard	Wrap SAP / plant APIs once
Personal layer	Cowork and Agent Builder on top

04

Governance and the Road Ahead

Running Fabric and AI safely, and a roadmap for SATORP

- Security
- Purview
- Capacity
- Use cases
- Roadmap
- Quality gate

Governance and Security in Fabric

The same principles from Day 1, applied to data.

01 Workspace roles

Admin, Member, Contributor and Viewer control who builds and who reads.

02 Item and OneLake security

Share single items; restrict folders and tables down to rows and columns.

03 Sensitivity labels

Microsoft Purview labels flow with data from the lakehouse into Power BI and exports.

04 Lineage and catalog

See where data came from and who uses it in the OneLake catalog.

05 Endorsement

Mark trusted items as Promoted or Certified so people pick the right one.

06 Deployment pipelines & Git

Move content from Dev to Test to Prod with version control.

Govern the data once in Fabric, and every report and agent built on it inherits the rules.

Capacity: How Fabric Is Paid For

You buy a capacity size (F-SKU). Every workload in its workspaces shares it.

01 F-SKUs

Capacities from F2 upwards. Bigger SKUs give more compute units for more users and workloads.

02 Shared pool

Pipelines, notebooks, warehouses, Power BI and Copilot all draw from the same capacity.

03 Capacity Metrics app

See compute-unit consumption by item and spot what is driving load.

04 Right-size and schedule

Start small, monitor, scale up or pause when not needed.

FOR AI FEATURES

- Copilot and data agents need F2+
- They consume capacity like other workloads
- Monitor after go-live

Capacity sizing is a cost-control lever.

Eight SATORP Use Cases, Mapped to Platforms

Example use cases for planning. Data sources and owners to be confirmed with SATORP teams.

#	Use case	Platform	Why
1	Shift handover & meeting summariser	Microsoft 365 Copilot	Personal productivity over M365 content
2	Team procedures helper	M365 Copilot Agent Builder	Small audience, documents only
3	HSE & HR Policy Assistant (Arabic + English)	Copilot Studio	Company-wide, governed, analytics
4	SATORP Public Q&A Assistant	Copilot Studio standard harness	Public website, designed menus
5	Maintenance request intake	Copilot Studio Workflow	Extract/Classify, approvals
6	Reliability & production KPI Q&A	Fabric data agent	Structured numbers in OneLake
7	Engineering & turnaround planning agent	Microsoft Foundry	Private SAP / OpenText, strict evaluation
8	Contractor permit & document check	Copilot Studio → Foundry if needed	Start low-code, scale only if required

A Suggested Roadmap in Three Phases

Start with quick wins to build trust, then scale with governance in place.

01
Quick wins

02
Governed enterprise agents

03
Engineering-grade AI

0–3 months

- Shift handover summaries (UC1)
- Team procedures helper (UC2)
- Public Q&A Assistant (UC4)
- Goal: adoption and trust, low risk

3–6 months

- HSE & HR Policy Assistant (UC3)
- Maintenance intake (UC5)
- Reliability KPI data agent (UC6)
- Goal: DLP, ALM and evaluation gates

6–12 months

- Engineering & turnaround agent (UC7)
- Permit check at scale (UC8)
- Goal: deep SAP / OpenText integration

Make it stick

- A small Centre of Excellence
- Ten intake questions for every idea
- An agent register with owners
- A decision record per use case

Start at the simplest platform that meets the need. Move down only when a requirement forces you to.

Anti-Patterns and What to Do Instead

Lessons from the decision guide that save months of rework.

	Anti-pattern	Instead
1	✕ Starting in Foundry “because it’s more powerful”	✓ Start at the simplest platform that meets the need
2	✕ Copying the same SOPs into five different agents	✓ One governed knowledge source, reused
3	✕ A Fabric data agent over raw, unnamed tables	✓ Curate tables, descriptions and example queries first
4	✕ Publishing without a test set	✓ Apply the minimum quality gate
5	✕ Org-wide agents built in personal Agent Builder	✓ Move to Copilot Studio with ALM, DLP and analytics
6	✕ Direct database access from agents to SAP HANA	✓ SAP APIs through APIM or MCP with SAP authorisations
7	✕ No owner after go-live	✓ Named business and technical owner, monthly review

Five Days, One Picture

What you now know, and can build.

Day 1
Foundations and M365 Copilot agents

Day 2
Copilot Studio agents, workflows, apps

Day 3
Foundry models, evaluation, security

Day 4
Clean data, RAG, Power BI

Day 5
Fabric, Copilot, IQ, data agents

The five rules to take back

- Start simple; move down only when forced
- Ground every answer in approved, owned sources
- Agents recommend; people decide on risk
- No test set, no publish
- Every agent has a named owner

Your next steps

- Pick one quick-win use case per team
- Fill in a decision record for it
- Write 20–50 real test questions
- Book a review with the Centre of Excellence
- Keep the lab repository as your reference

PROGRAM COMPLETE

Thank You, SATORP

Microsoft Data & AI Readiness for SATORP · Delivered by CodeSizzler

NEXT

Questions, feedback and next steps. We are happy to help SATORP teams take the first quick-win use cases from idea to production.

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